Gallatin-Sumner County Geoeconomic Growth Risk Monitor

A Python-based local economic intelligence model measuring growth pressure, construction risk, infrastructure strain, development friction, household vulnerability, industry exposure, and geoeconomic shock sensitivity in Gallatin and Sumner County, Tennessee.

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Executive Summary

This report develops a local economic intelligence model for Gallatin and Sumner County, Tennessee. The model evaluates whether rapid local growth could become more fragile if development pressure intersects with geopolitical or macroeconomic shocks affecting construction costs, fuel prices, financing costs, logistics costs, housing affordability, infrastructure strain, household vulnerability, industry exposure, and supply chains.

This version is a fully data-driven local risk monitor. Growth pressure, construction shock exposure, housing affordability pressure, infrastructure strain, development friction, geoeconomic industry exposure, household vulnerability, and local resilience are all active model modules. No composite risk component is currently treated as a placeholder.

Because the adjusted score remains in the Elevated Risk band even after the resilience credit, the model suggests that local resilience capacity helps reduce risk but does not fully offset growth, infrastructure, housing, development, and industry-exposure pressure.

The Overheated Growth scenario matters because it tests whether rapid development could outpace housing affordability, road capacity, utilities, planning capacity, and public-service absorption.

Metric	Latest Value
Composite Risk Score	69.1
Local Resilience Offset	55.0
Resilience Credit Applied	8.2
Adjusted Geoeconomic Growth Risk Score	60.9
Adjusted Risk Signal	Elevated Risk
Highest-Risk Scenario	Overheated Growth
Highest-Risk Scenario Score	64.7
Highest-Risk Scenario Delta vs. Baseline	+3.8
Model Maturity	Fully Data-Driven
Data Quality Flag	Data-Driven Components Available
Live / Placeholder Components	7 live / 0 placeholder
Placeholder Share	0.0%

Key Findings

#	Finding
1	The current composite risk score is 69.1, while the adjusted risk score is 60.9 after applying a capped resilience credit of 8.2.
2	Growth pressure remains one of the most important weighted local risk signals, with a latest score of 85.4 and a 20% composite weight.
3	Housing affordability pressure is a live module, with a latest score of 49.5 and signal of Moderate Risk.
4	Infrastructure strain is a live module, with a latest score of 69.9 and signal of Elevated Risk.
5	Development friction is a live module, with a latest score of 84.2 and signal of High Risk.
6	Geoeconomic industry exposure is a live module, with a latest score of 98.5 and signal of High Risk.
7	Household vulnerability is a live module, with a latest score of 55.9 and signal of Elevated Risk.
8	Construction shock exposure is currently 40.6. The latest short-term momentum signal is: Rapidly Rising.
9	The highest-risk scenario is Overheated Growth, with an adjusted scenario risk score of 64.7.
10	Local resilience evidence confidence is currently classified as Strong Evidence, based on curated source notes.

Latest Module Scores

Module	Latest Score	Signal / Context
Construction Shock Exposure	40.6	Moderate Risk | Momentum: Rapidly Rising
Growth Pressure	85.4	High Risk | Partial Economic Data
Housing Affordability Pressure	49.5	Moderate Risk | Strong ACS Coverage | ACS 2024
Infrastructure Strain	69.9	Elevated Risk | Strong ACS Commute Coverage | Local signal: Strong Local Signal
Development Friction	84.2	High Risk | Source signal: Very Strong Source Signal
Geoeconomic Industry Exposure	98.5	High Risk | Very Strong Source Signal | ACS 2024
Household Vulnerability	55.9	Elevated Risk | Strong ACS Household Coverage | Local signal: Very Strong Local Signal
Local Resilience Offset	55.0	Strong Resilience Offset | Confidence: Strong Evidence

How to Read the Results

The model uses a 0-100 scoring framework. Higher values indicate more pressure, not a guaranteed outcome. The scores should be read as an intelligence dashboard: they show where local growth, affordability, infrastructure, development, industry, and household stress signals are concentrated and how those signals move over time.

Score Range	Risk Band	How to Interpret
0-24.9	Low Risk	Limited current pressure; monitor for early movement.
25.0-49.9	Moderate Risk	Manageable pressure; useful as a watchlist signal.
50.0-74.9	Elevated Risk	Meaningful pressure; deserves regular monitoring and explanation.
75.0-100	High Risk	Material pressure; likely a priority risk channel or policy focus area.

Result	Current Reading	How to Use It
Composite Risk Score	69.1 / Elevated Risk	Read this as the gross local risk pressure before resilience is applied.
Adjusted Risk Score	60.9 / Elevated Risk	Use this as the headline result because it reflects both risk pressure and the capped local resilience credit.
Resilience Credit	8.2	Shows how much local response capacity reduced the gross composite score. It is capped so it cannot erase high underlying risk.
Highest-Risk Scenario	Overheated Growth (64.7, +3.8)	Identifies the stress case that would most increase local risk relative to the adjusted baseline.
Breach Monitor	Elevated Alert	Summarizes whether any metrics crossed configured warning thresholds.
Model Maturity	Fully Data-Driven; 7 live / 0 placeholder; 0.0% placeholder share	Shows whether the model is driven by live data modules or placeholder assumptions.

A practical reading of the report should start with the adjusted risk score, then move to the component table to identify which modules are driving the result, then review scenarios, sensitivity analysis, and threshold breaches to determine whether the conclusion is robust or dependent on one assumption.

Metric Measurement Guide

This section explains the main metrics used in the report. Each metric is designed to translate raw economic, demographic, source-note, and market data into a repeatable 0-100 monitoring signal. The scores are decision-support indicators, not probabilities or official forecasts.

Core Result Metrics

Metric	What It Measures	How It Can Be Used
Composite Risk Score	Weighted average of the seven live risk components before resilience adjustment.	Use as the broadest measure of local geoeconomic growth pressure.
Adjusted Risk Score	Composite score minus a capped resilience credit.	Use as the headline risk reading because it includes local response capacity.
Local Resilience Offset	Evidence of utility coordination, public works capacity, flood mitigation, public safety response, and capital execution.	Use to measure local response capacity, but only as a capped risk reducer.

Risk Component Metrics

Metric	What It Measures	How It Can Be Used
Growth Pressure	Local growth, labor-market, and source-note signals tied to development demand.	Use to monitor whether growth is becoming difficult to absorb.
Housing Affordability	ACS income, home value, rent, cost-burden, and mortgage-rate pressure.	Use to assess whether households can absorb local housing and financing costs.
Construction Shock Exposure	FRED-based materials, fuel/energy, and financing cost pressure.	Use to monitor external cost shocks that can delay or reprice local projects.
Infrastructure Strain	ACS commute patterns, vehicle dependence, long-commute exposure, and infrastructure source notes.	Use to assess pressure on roads, transportation systems, utilities, and public works capacity.
Development Friction	Local source evidence on zoning, planning, permitting, development pipeline, and public notices.	Use to monitor whether policy, process, or development complexity is increasing.
Geoeconomic Industry Exposure	Employment-exposure indicators plus local evidence for manufacturing, logistics, construction, retail, and trade-sensitive sectors.	Use to understand how external shocks could transmit through the local economic base.
Household Vulnerability	ACS poverty, low-income exposure, unemployment, housing burden, commute burden, and local vulnerability notes.	Use to identify whether households may be more sensitive to inflation, housing, job, or transportation shocks.

Monitoring and Validation Metrics

Metric	What It Measures	How It Can Be Used
Scenario Scores	Stress-test outputs under managed growth, overheated growth, construction shock, and rezoning constraint cases.	Use to compare which future condition creates the greatest local risk pressure.
Sensitivity Analysis	Alternative weighting cases that emphasize different risk channels.	Use to test whether the report conclusion remains stable under different assumptions.
Risk Threshold Breaches	Configured warning thresholds for key risk, scenario, resilience, and model-quality metrics.	Use as an alert layer to prioritize which metrics need attention first.
Change Detection	Current model values compared with the prior saved run snapshot.	Use to monitor whether risk pressure is improving, worsening, or stable over time.

The metric values should not be interpreted as probabilities. They are normalized monitoring scores intended to support comparison across risk channels, identify trend changes, and guide where deeper local research or policy review may be needed.

Factual Interpretation and Validation Notes

This report should be read as a repeatable local intelligence model rather than an official government forecast or a deterministic prediction. The scores should be interpreted as portfolio-style intelligence indicators, not official public-sector risk ratings. The analysis combines public economic data, ACS demographic indicators, FRED market indicators, and curated local source notes. Where the report uses source-note evidence, those notes strengthen local context and confidence but do not replace official numeric datasets.

Area	What to Check	Why It Matters
ACS Indicators	Confirm geography, year, dataset, variable definitions, and whether values are counts, percentages, or normalized model inputs.	Prevents unit errors and avoids overstating demographic or employment totals.
FRED Indicators	Confirm latest observation dates and whether a series is daily, weekly, monthly, or annual.	Prevents comparing stale or differently timed macro indicators as if they were equivalent.
Local Source Notes	Confirm source date, jurisdiction, topic, and risk channel before treating a note as evidence.	Keeps qualitative signals grounded in documented local sources.
Composite Scores	Read scores as normalized monitoring indicators, not probabilities or official risk ratings.	Prevents overclaiming what the model can measure.
Scenario Results	Treat scenario scores as assumption-based stress tests, not forecasts.	Keeps scenario analysis framed as decision support rather than prediction.

Recommended proofread pass before publication: verify that every numeric value has a clear source, that source-note claims are described as qualitative evidence, and that all forward-looking language uses terms such as could, may, scenario, pressure, or exposure rather than certainty.

Change Detection Summary

No material changes were detected versus the prior run.

Metric	Current	Previous	Delta	Direction	Materiality
Composite Risk Score	69.1	69.1	+0.0	Stable	Minimal
Adjusted Geoeconomic Growth Risk Score	60.9	60.9	+0.0	Stable	Minimal
Growth Pressure Score	85.4	85.4	+0.0	Stable	Minimal
Housing Affordability Pressure Score	49.5	49.5	+0.0	Stable	Minimal
Construction Shock Score	40.6	40.6	+0.0	Stable	Minimal
Fuel Energy Shock Score	74.1	74.1	+0.0	Stable	Minimal
Infrastructure Strain Score	69.9	69.9	+0.0	Stable	Minimal
Development Friction Score	84.2	84.2	+0.0	Stable	Minimal
Geoeconomic Industry Exposure Score	98.5	98.5	+0.0	Stable	Minimal
Household Vulnerability Score	55.9	55.9	+0.0	Stable	Minimal
Local Resilience Offset	55.0	55.0	+0.0	Stable	Minimal
Highest Risk Scenario Score	64.7	64.7	+0.0	Stable	Minimal
Highest Risk Scenario	Overheated Growth	Overheated Growth	N/A	Stable	Minimal

Change detection compares the current model run against the prior saved snapshot. The first run establishes the baseline for future comparisons.

Sensitivity Analysis

The sensitivity analysis tests whether the model conclusion changes under alternative weighting assumptions. This is useful for checking whether the conclusion is driven by one model component or remains stable across alternative risk-emphasis cases.

Metric	Value
Sensitivity Stability	Stable
Minimum Adjusted Risk Score	55.4
Maximum Adjusted Risk Score	63.5
Adjusted Risk Score Range	8.1
Lowest-Risk Weighting Case	Construction-Shock-Heavy Case
Highest-Risk Weighting Case	Infrastructure / Friction Heavy Case

Weighting Case	Composite	Adjusted	Signal
Baseline Weights	69.1	60.9	Elevated Risk
Growth-Heavy Case	71.1	62.8	Elevated Risk
Construction-Shock-Heavy Case	63.6	55.4	Elevated Risk
Infrastructure / Friction Heavy Case	71.7	63.5	Elevated Risk
Household-Vulnerability-Heavy Case	65.0	56.7	Elevated Risk
Equal-Weight Case	69.1	60.9	Elevated Risk

A stable sensitivity result means the model's broad risk conclusion is not overly dependent on one weighting framework.

Risk Threshold Breach Monitor

High-severity thresholds are active in: Growth Pressure. Active risk thresholds are present in: Composite Risk, Growth Pressure, Scenario Stress. The highest-risk scenario is Overheated Growth, with a delta of +3.8 versus the adjusted

baseline. Overall breach monitor status: Elevated Alert.

Metric	Value
Alert Level	Elevated Alert
Total Active Breaches	4
High-Severity Breaches	1
Model Quality Breaches	0
Top Breach Category	Growth Pressure
Top Breach Description	Growth pressure is high, suggesting local absorption risk.
Breached Categories	Composite Risk, Growth Pressure, Scenario Stress

Category	Metric	Value	Threshold	Severity
Growth Pressure	Growth Pressure Score	85.4	75.0	High
Composite Risk	Composite Risk Score	69.1	50.0	Medium
Scenario Stress	Highest Scenario Score	64.7	50.0	Medium
Composite Risk	Adjusted Geoeconomic Growth Risk Score	60.9	50.0	Medium

The breach monitor converts model outputs into an alert-style layer by tracking whether key growth, construction, scenario, resilience, and model-quality thresholds are active.

How to Read the Component and Scenario Tables

Table	How to Read It
Composite Risk Component Summary	Focus first on components above 75, then components between 50 and 75. High components show the strongest current pressure channels.
Component Weight	The weight shows how much each module contributes to the composite score. A high score with a high weight has more impact on the headline result.
Signal	The signal converts a numeric score into a readable risk band: Low, Moderate, Elevated, or High.
Status	Live means the component is driven by a dedicated module or dataset. Placeholder would mean a temporary assumption, but the current model has no placeholder components.
Scenario Delta	Positive delta means the scenario increases risk versus the adjusted baseline. Negative delta means the scenario reduces risk versus the adjusted baseline.
Primary Risk Driver	This field explains which local channel is most responsible for each scenario result.

Composite Risk Component Summary

Component	Weight	Score	Signal	Status
Growth Pressure	20%	85.4	High Risk	Live
Housing Affordability Pressure	15%	49.5	Moderate Risk	Live
Construction Shock Exposure	15%	40.6	Moderate Risk	Live
Infrastructure Strain	15%	69.9	Elevated Risk	Live
Development Friction	15%	84.2	High Risk	Live
Geoeconomic Industry Exposure	10%	98.5	High Risk	Live
Household Vulnerability	10%	55.9	Elevated Risk	Live

Scenario Forecasts

The scenario model estimates how local growth risk could change under alternative growth, construction-cost, infrastructure, housing, and planning conditions. The delta column compares each scenario with the latest adjusted baseline risk score.

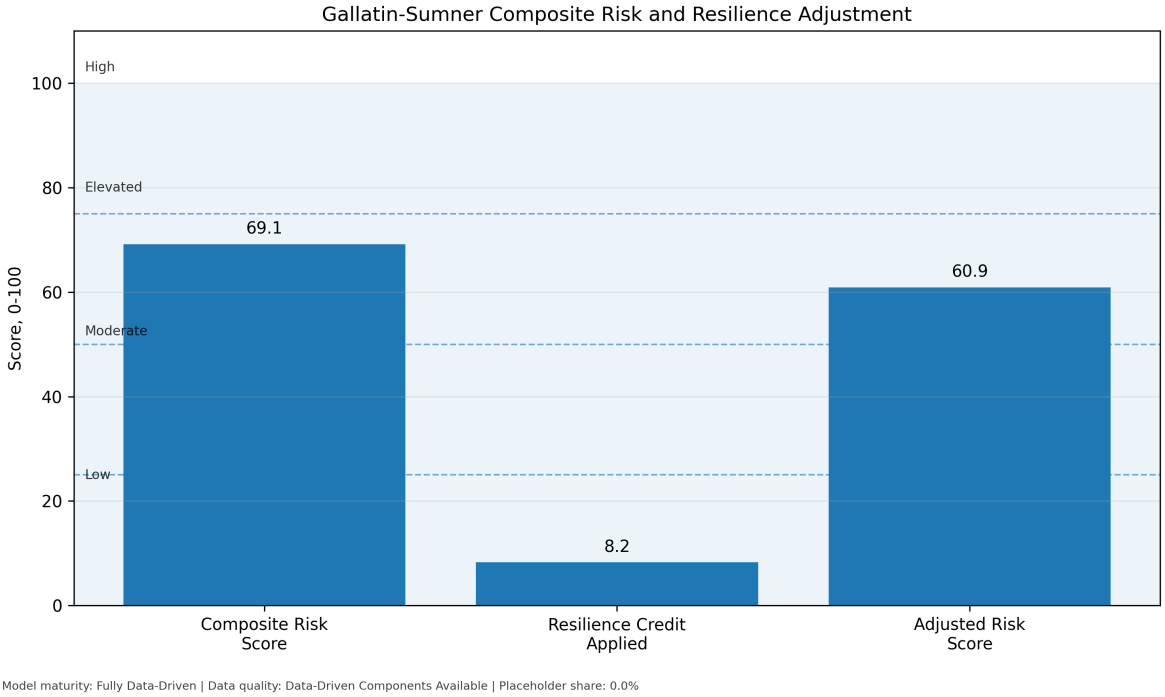
Scenario	Score	Delta	Signal	Primary Risk Driver
Overheated Growth	64.7	+3.8	Elevated Risk	Infrastructure, housing, and development pressure
Geopolitical Construction Cost Shock	63.6	+2.7	Elevated Risk	Materials, fuel, financing, and supply-chain costs
Growth Pause / Rezoning Constraint	55.1	-5.8	Elevated Risk	Rezoning uncertainty and housing supply constraints
Managed Growth	55.0	-5.9	Elevated Risk	Baseline growth absorption

Scenario	Interpretation Note
Overheated Growth	This is the main local stress case because it tests rapid growth against housing, road, utility, and planning capacity.
Geopolitical Construction Cost Shock	This scenario tests external geoeconomic shock exposure through materials, fuel, financing, and logistics costs.
Growth Pause / Rezoning Constraint	This scenario tests policy and planning friction, where lower growth may reduce some infrastructure pressure but increase housing supply risk.
Managed Growth	This scenario should be interpreted as a controlled baseline rather than a downside shock.

How to Read the Charts

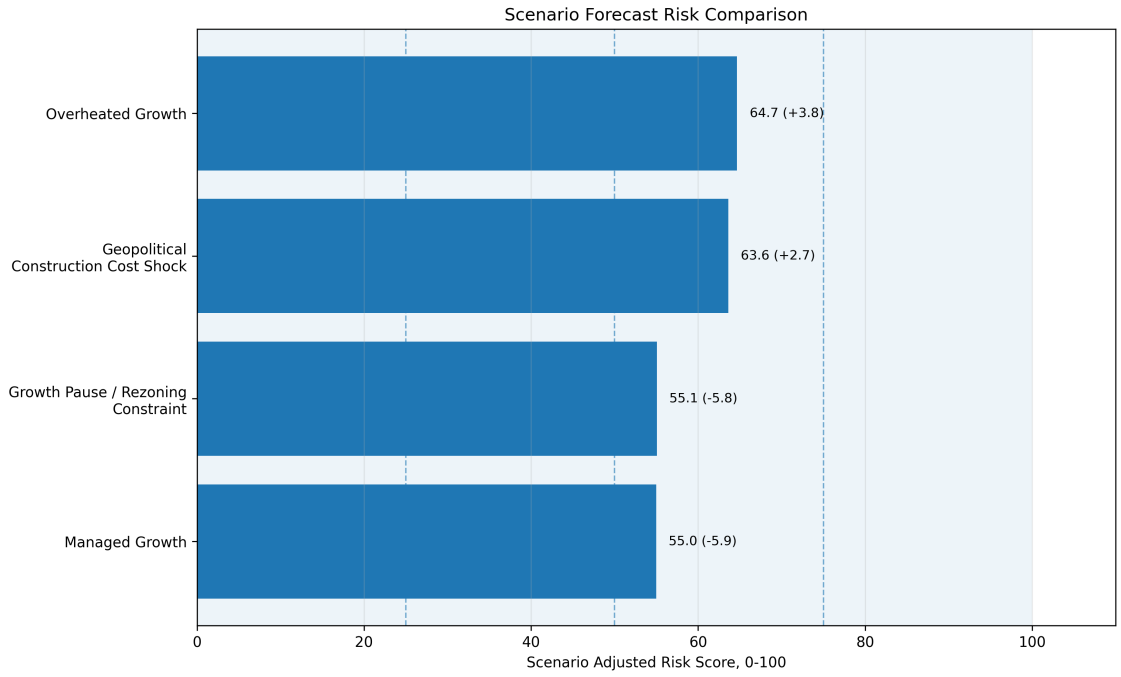
The charts translate the model outputs into a visual monitoring layer. Read each chart from left to right or top to bottom, then compare the chart signal with the numeric tables. The most useful pattern is not a single chart value, but whether multiple charts point to the same risk channel: growth absorption, affordability, infrastructure strain, cost shocks, planning friction, or resilience capacity.

Composite Risk Score and Resilience Adjustment



How to read this chart: the first bar shows the gross composite risk score before any offset, the middle bar shows the capped resilience credit applied, and the final bar shows the adjusted risk score. If the adjusted score remains elevated after the resilience credit, local capacity is helpful but not enough to eliminate the underlying growth, affordability, infrastructure, and industry-exposure pressure.

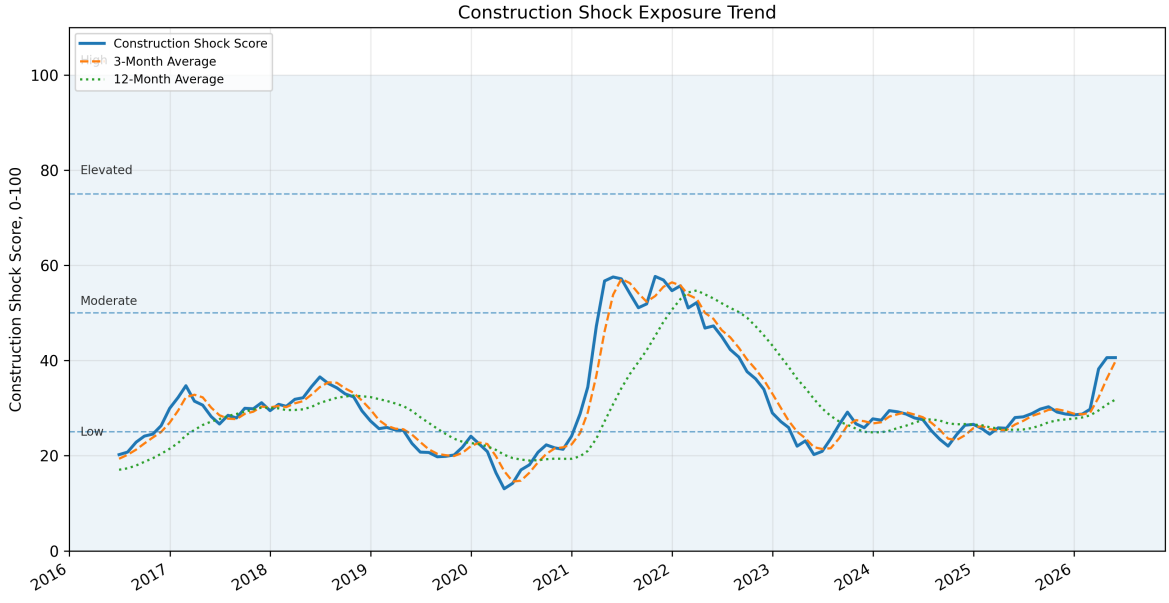
Scenario Forecast Comparison



Labels show score and delta vs. adjusted baseline where available. Model maturity: Fully Data-Driven | Data quality: Data-Driven Components Available

How to read this chart: each bar is a stress-case estimate of adjusted risk under different assumptions. The label in parentheses shows the change versus the current adjusted baseline. The highest bar identifies the scenario that would create the most pressure on the local system.

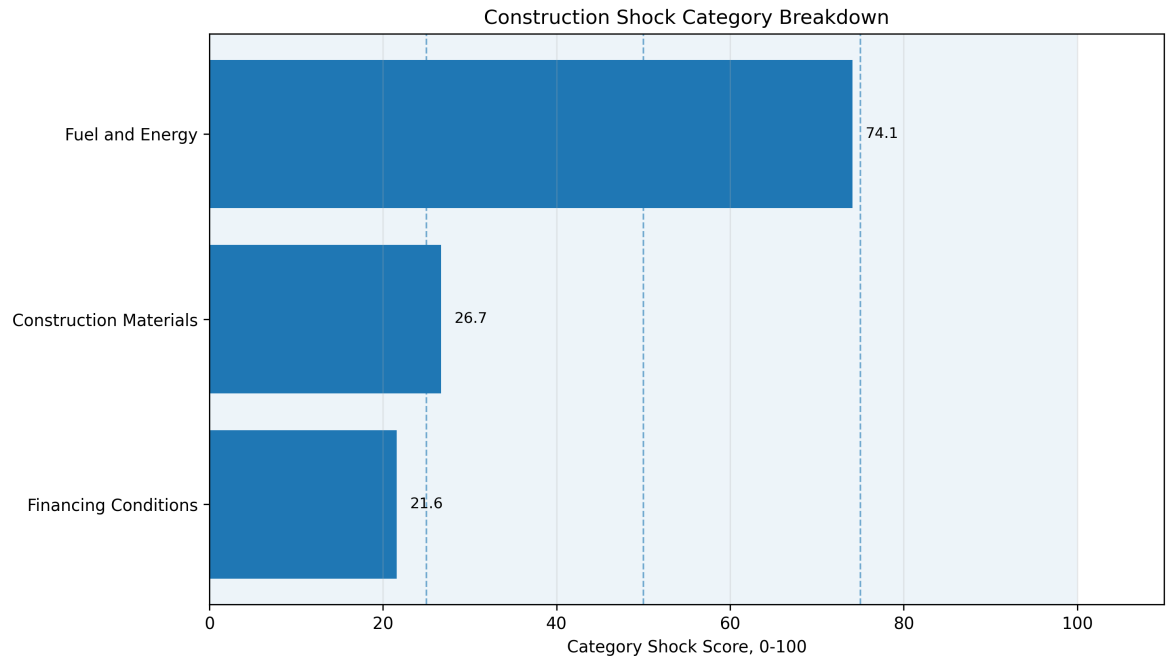
Construction Shock Trend



Construction shock score uses input-cost, fuel/energy, and financing indicators. Latest momentum: Rapidly Rising. Data quality: Strong Component Coverage.

How to read this chart: the line tracks construction-cost shock exposure through time, while the moving averages smooth short-term noise. Rising momentum indicates that materials, fuel, or financing conditions are becoming less favorable for local projects and development execution.

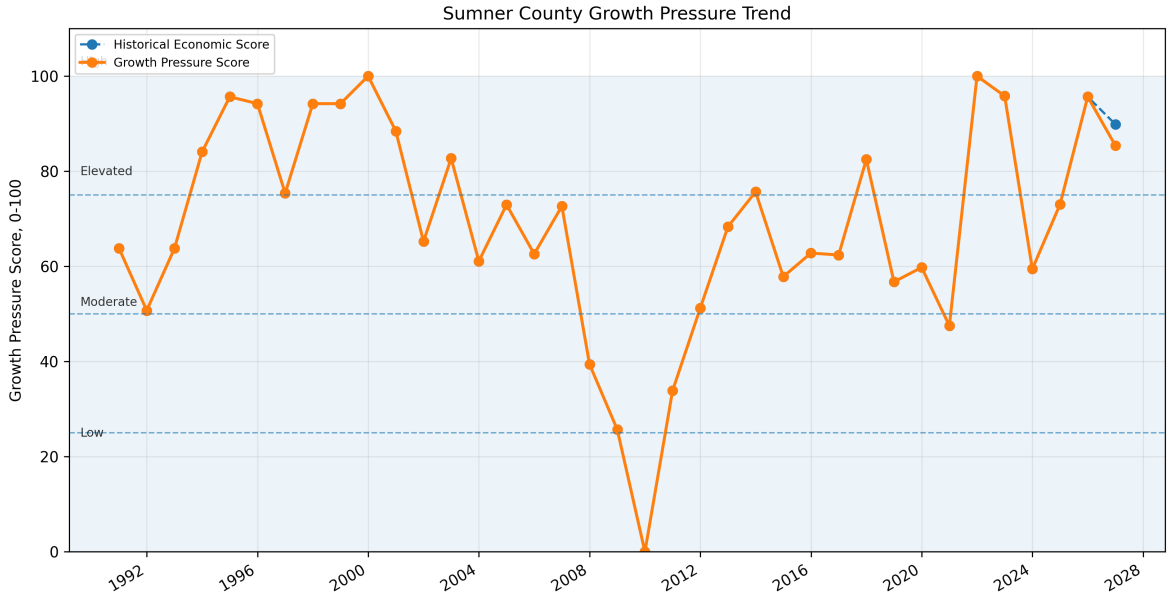
Construction Shock Category Breakdown



Category scores summarize materials, fuel/energy, and financing conditions.

How to read this chart: the bars separate the construction-shock score into materials, fuel and energy, and financing conditions. This helps identify whether current pressure is being driven by input costs, energy/logistics costs, or interest-rate conditions.

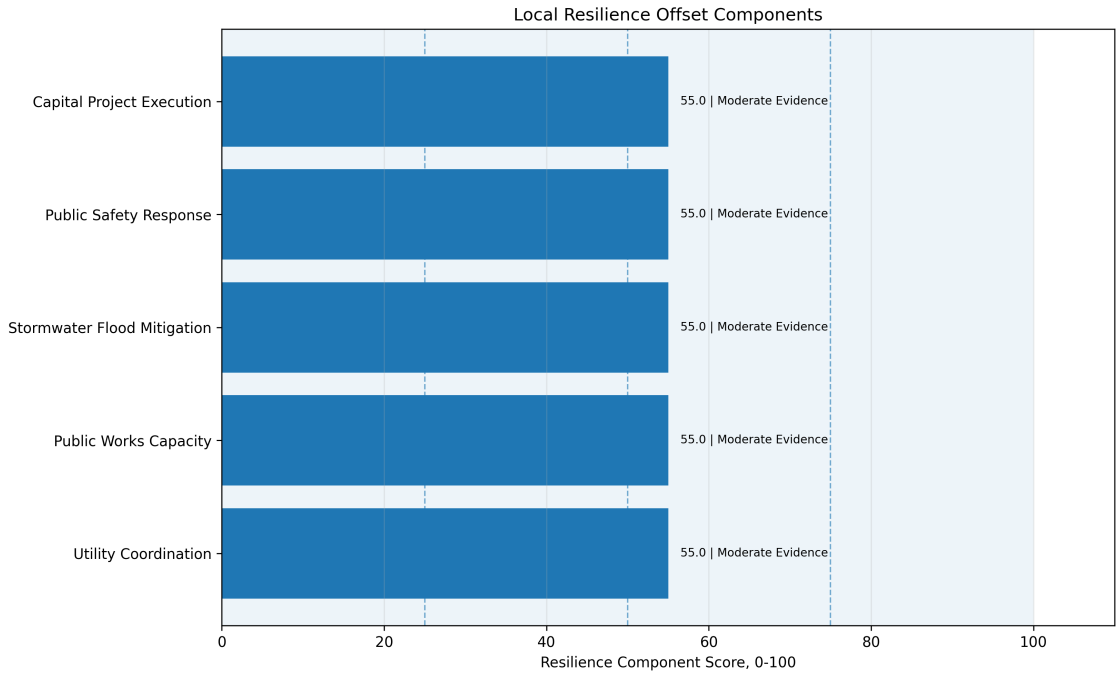
Growth Pressure Trend



Historical scores use economic data. Latest score may include current local-source signal. Local signal: Strong Qualitative Signal. Data quality: Partial Economic Data.

How to read this chart: the trend shows how local growth pressure has evolved over time. High or rising values suggest stronger absorption risk, meaning population, labor, development, and infrastructure demand may be moving faster than local systems can comfortably support.

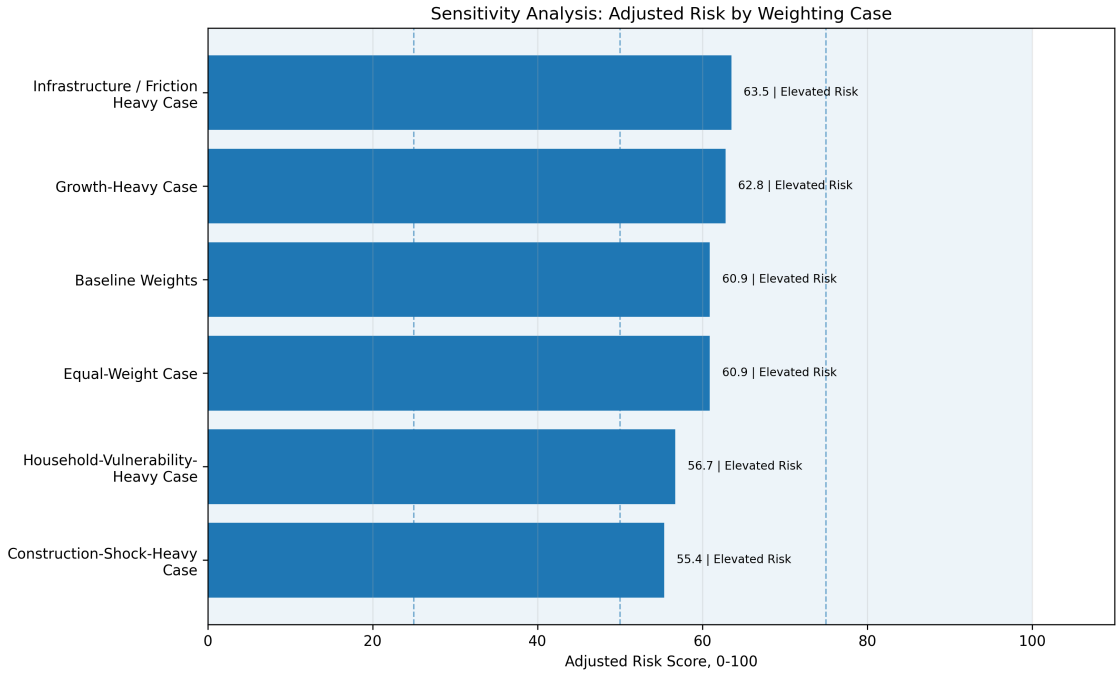
Local Resilience Offset Components



Resilience scores use conservative evidence tiers from curated local source notes.

How to read this chart: each bar shows a resilience evidence tier for a local capacity channel. Higher values indicate stronger evidence of response capacity, but these scores are converted into a capped credit so resilience reduces risk without overpowering the core risk signals.

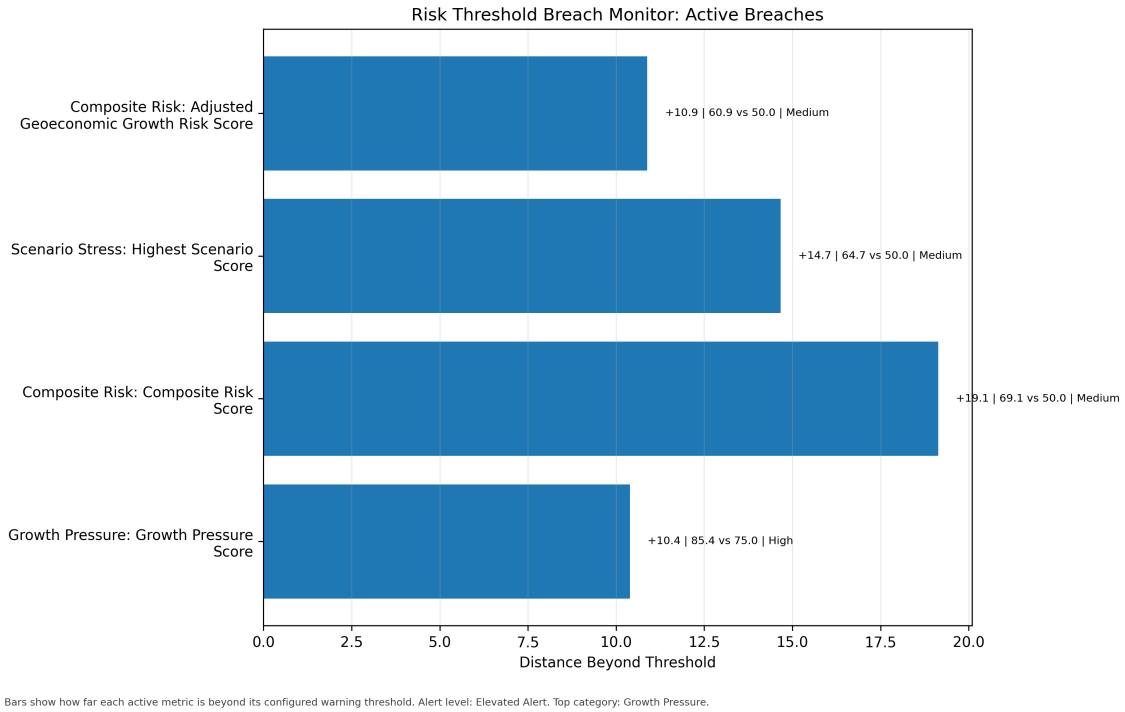
Sensitivity Analysis



Sensitivity analysis tests whether the model conclusion changes under alternative weighting assumptions. Stability: Stable. Score range: 8.1.

How to read this chart: each bar recalculates adjusted risk under a different weighting case. If all bars remain in the same broad risk band, the conclusion is stable and not overly dependent on one weighting framework.

Risk Threshold Breach Monitor



How to read this chart: each bar shows how far an active metric is beyond its configured warning threshold. Longer bars indicate larger threshold breaches and therefore clearer monitoring priorities.

Housing Affordability Detail

Metric	Value
Geography	Sumner County, Tennessee
ACS Year	2024
ACS Dataset	acs/acs5
Median Household Income	$90,301
Median Home Value	$393,100
Median Gross Rent	$1,424
Home Value to Income	4.35x
Rent to Income	18.9%
Renter Cost-Burden Share	46.0%
Owner Cost-Burden Share	17.1%
Mortgage Rate	6.4%
Housing Affordability Pressure Score	49.5
Housing Affordability Signal	Moderate Risk
Data Quality	Strong ACS Coverage

Housing Component Scores

Component	Weight	Score	Signal
Home Value to Income	30%	58.8	Elevated Risk
Rent to Income	20%	15.7	Low Risk
Renter Cost Burden	20%	88.6	High Risk
Owner Cost Burden	15%	5.9	Low Risk
Mortgage Rate Pressure	15%	67.2	Elevated Risk

The housing affordability module uses ACS income, home value, rent, cost-burden, and mortgage-rate indicators to estimate household affordability pressure.

Infrastructure Strain Detail

Metric	Value
Geography	Sumner County, Tennessee
ACS Year	2024
ACS Dataset	acs/acs5
Mean Travel Time	34.4 minutes
Long Commute Share	24.7%
Drive-Alone Share	74.0%
Carpool Share	7.7%
Public Transport Share	0.1%
Worked From Home Share	16.2%
Local Infrastructure Notes	23

Metric	Value
Public Works / Road Notes	12
Source Signal Confidence	Strong Local Signal
Infrastructure Strain Score	69.9
Infrastructure Strain Signal	Elevated Risk
Data Quality	Strong ACS Commute Coverage

Infrastructure Component Scores

Component	Weight	Score	Signal
Mean Commute Time Pressure	30%	77.7	High Risk
Long Commute Share	20%	65.5	Elevated Risk
Vehicle Dependence	20%	39.9	Moderate Risk
Local Infrastructure Note Signal	15%	85.0	High Risk
Public Works / Road Pressure	15%	85.0	High Risk

The infrastructure strain module uses ACS commute and transportation indicators plus curated local source notes related to roads, public works, utilities, stormwater, and development pressure.

Development Friction Detail

Metric	Value
Development Friction Score	84.2
Development Friction Signal	High Risk
Source Confidence Score	90.0
Source Confidence Label	Very Strong Source Signal
Total Relevant Notes	31
Unique Sources	9
Source Type Count	7
Zoning / Code Notes	3
Planning / Permitting Notes	5
Development Pipeline Notes	19
Public Notice / Agenda Notes	4
Data Quality	Very Strong Source Signal

Development Friction Component Scores

Component	Weight	Score	Signal
Zoning / Code Update Signal	30%	70.6	Elevated Risk
Planning / Permitting Signal	25%	90.0	High Risk
Development Pipeline Pressure	20%	90.0	High Risk
Public Notice / Agenda Activity	15%	90.0	High Risk
Source Confidence	10%	90.0	High Risk

The development friction module uses curated local source notes to estimate pressure from zoning-code updates, planning/permitting activity, public notice and agenda activity, and the development pipeline.

Geoeconomic Industry Exposure Detail

Metric	Value
Geography	Sumner County, Tennessee
ACS Year	2024
ACS Dataset	acs/acs5/profile
ACS Industry Base Count	577.0
Manufacturing Share	5.3%
Logistics / Transportation Share	3.4%
Construction Share	17.8%
Retail Trade Share	11.0%
Wholesale Trade Share	23.3%
Trade-Sensitive Share	60.8%
Manufacturing Source Notes	4
Logistics Source Notes	7
Trade-Sensitive Source Notes	11
Source Confidence	Very Strong Source Signal
Industry Exposure Score	98.5
Industry Exposure Signal	High Risk
Data Quality	Very Strong Source Signal

Industry Exposure Component Scores

Component	Weight	Score	Signal
Manufacturing Exposure	25%	100.0	High Risk
Logistics / Transportation Exposure	20%	100.0	High Risk
Construction / Development Exposure	20%	100.0	High Risk
Retail / Local Demand Exposure	15%	100.0	High Risk
Trade-Sensitive Growth Signal	10%	90.0	High Risk
Source Confidence	10%	95.0	High Risk

The geoeconomic industry exposure module combines employment-exposure indicators with curated local source notes to estimate exposure to manufacturing, logistics, construction, retail, and trade-sensitive growth channels. Counts and percentages should be interpreted within the module's ACS variable definitions and validated before use as official employment totals.

Household Vulnerability Detail

Metric	Value
Geography	Sumner County, Tennessee
ACS Year	2024
ACS Dataset	acs/acs5/profile
Households Total	78582.0

Metric	Value
Median Household Income	$90,301
Per Capita Income	$44,761
Poverty Rate	8.8%
Unemployment Rate	3.6%
Low-Income Household Share	26.3%
Housing Cost-Burden Share	29.3%
Mean Travel Time	29.0 minutes
Local Vulnerability Notes	16
Local Vulnerability Signal	Very Strong Local Vulnerability Signal
Household Vulnerability Score	55.9
Household Vulnerability Signal	Elevated Risk
Data Quality	Strong ACS Household Coverage

Household Vulnerability Component Scores

Component	Weight	Score	Signal
Poverty Exposure	25%	17.5	Low Risk
Low-Income Household Exposure	20%	100.0	High Risk
Unemployment Exposure	18%	10.0	Low Risk
Housing Cost Burden	15%	100.0	High Risk
Commute Burden	10%	38.9	Moderate Risk
Local Vulnerability Source Signal	7%	90.0	High Risk
Data Quality	5%	90.0	High Risk

Household Vulnerability Source Signal

Metric	Value
Total Relevant Notes	16
Unique Sources	1
Source Type Count	1
Local Vulnerability Source Signal Score	90.0
Local Vulnerability Source Signal Label	Very Strong Local Vulnerability Signal

The household vulnerability module uses ACS poverty, income, unemployment, housing cost burden, and commute indicators, plus curated local source notes related to affordability, vulnerable residents, transportation cost exposure, and social-service capacity.

Construction Shock Detail

Category Breakdown

Category	Score	Signal
Construction Materials	26.7	Moderate Risk
Fuel and Energy	74.1	Elevated Risk
Financing Conditions	21.6	Low Risk

Latest Component Scores

Component	Category	Score	YoY %	Weight
Construction Materials PPI	Materials	30.6	4.6	20%
Lumber PPI	Materials	27.9	3.9	10%
Cement PPI	Materials	20.4	-0.5	10%
Steel Mill Products PPI	Materials	23.9	5.3	10%
Diesel Price	Fuel Energy	88.4	60.5	15%
Retail Gasoline Price	Fuel Energy	81.8	42.1	10%
WTI Crude Oil	Fuel Energy	45.0	65.2	10%
30-Year Mortgage Rate	Financing	23.4	-7.7	15%
Federal Funds Rate	Financing	18.8	-15.9	10%

Local Resilience Evidence Detail

Component	Score	Confidence	Notes	Sources
Utility Coordination	55.0	Moderate Evidence	10	6
Public Works Capacity	55.0	Moderate Evidence	5	3
Stormwater Flood Mitigation	55.0	Moderate Evidence	2	2
Public Safety Response	55.0	Moderate Evidence	3	2
Capital Project Execution	55.0	Moderate Evidence	8	5

The resilience module uses conservative evidence tiers so a small number of keyword matches does not automatically create maximum resilience scores.

Methodology

The model uses a 0-100 scoring framework. Higher values indicate higher risk pressure. The current version is fully data-driven across all seven composite components: growth pressure, housing affordability, construction shock exposure, infrastructure strain, development friction, geoeconomic industry exposure, and household vulnerability. The Local Resilience Offset is handled separately as a capped risk-reduction credit.

Component	Model Role
Growth Pressure	Measures GDP growth, labor-market tightness, and current local growth/development signals.
Housing Affordability	Measures income, home value, rent, cost-burden, and mortgage-rate pressure.
Construction Shock Exposure	Measures sensitivity to materials, fuel, energy, and financing costs.
Infrastructure Strain	Measures commute pressure, vehicle dependence, long-commute exposure, and local infrastructure-note pressure.
Development Friction	Measures zoning, planning, permitting, development-pipeline, and public-notice pressure.
Geoeconomic Industry Exposure	Measures exposure to manufacturing, logistics, construction, retail, and trade-sensitive growth channels using employment-exposure indicators and local source evidence.
Household Vulnerability	Measures poverty, low-income exposure, unemployment, housing cost burden, commute burden, and local vulnerability evidence.
Local Resilience Offset	Measures utility coordination, public works capacity, flood mitigation, public safety response, and project execution.
Capped Resilience Credit	Converts resilience into a capped risk reduction so resilience does not overpower risk pressure.
Scenario Forecasts	Stress-tests the model under managed growth, overheated growth, construction cost shock, and rezoning-constraint cases.
Change Detection	Compares the current run with the previous run snapshot to support repeatable monitoring.
Sensitivity Analysis	Tests whether the model conclusion changes under alternative weighting assumptions.
Risk Threshold Breach Monitor	Tracks whether key model indicators cross material warning thresholds.

Model Caveats and Next Improvements

The current model maturity level is **Fully Data-Driven**. The composite model currently uses 0 placeholder components, representing approximately 0.0% of the composite component count. Future work should focus on validation, automation, source expansion, and dashboard delivery rather than replacing placeholder components.

Future Enhancement	Purpose
Streamlit Dashboard	Convert the PDF workflow into an interactive local intelligence dashboard.
Source Evidence Tracker	Track source age, source type, jurisdiction, confidence, and which model components each source supports.
Policy Activity Tracker	Convert city/county meeting records, public notices, zoning actions, and planning agendas into a structured policy dataset.
Forecast Validation	Back-test scenario assumptions and compare model trends against future housing, permit, commute, labor, and industry data.
Document Center Integration	Use the City of Gallatin Document Center as a repeatable source hub for planning, zoning, infrastructure, and policy evidence.

Curated Local Source Notes

date	source_name	topic	risk_channel
2026	Gallatin Gets It	Economic growth narrative	growth_pressure
2026	Gallatin Gets It	Developable sites	growth_pressure
2026	Gallatin Gets It Location and Infrastructure	Road investment	infrastructure_resilience
2026	Gallatin Gets It Location and Infrastructure	Public utilities	local_resilience
2026	Gallatin Gets It High Tech Manufacturing	Manufacturing investment	geoeconomic_exposure
2026	Gallatin Gets It High Tech Manufacturing	Advanced manufacturing	geoeconomic_exposure
2026	Gallatin Gets It High Tech Manufacturing	Distribution expansion	logistics_supply_chain_exposure
2018	City of Gallatin Strategic Plan	Growth and development	growth_pressure
2018	City of Gallatin Strategic Plan	Key issues and trends	strategic_context
2018	City of Gallatin Strategic Plan	Transportation and mobility	infrastructure_strain
2018	City of Gallatin Strategic Plan	Affordable living	household_vulnerability
2018	City of Gallatin Strategic Plan	Resilient and diversified economy	geoeconomic_exposure
2018	City of Gallatin Strategic Plan	Purposeful planned growth	development_friction
2018	City of Gallatin Strategic Plan	Community vulnerability	household_vulnerability
2026	City of Gallatin Document Center	Planning and development records	development_friction

date	source_name	topic	risk_channel
2026	City of Gallatin Document Center	Engineering and stormwater records	infrastructure_strain

Note: Source notes are used as qualitative local indicators and are not treated as official numeric datasets. They are used to strengthen context and source confidence around local planning, infrastructure, affordability, resilience, and policy evidence.